Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Kosslyn, Stephen M. Ph.D.		
Project Title:	Quick Assessment of Basic Cognitive Function: Blood Pressure Cuffs for the Mind		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	02138-3846	**Congressional District:**	8
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
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Contact Monitor:		**Contact Phone:**	
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Performance Goal No.:			
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	The goal of this project is to develop a set of brief performance tests (the MiniCog Quick Assessment Battery, or MQAB) and a method for administering them (the MiniCog application) on a handheld device or PDA. The performance tests have been selected from among standard tasks in the psychological literature, and tap basic cognitive abilities necessary for performing complex operations (such as extravehicular activity—EVA—or many other tasks that astronauts have to complete) or that have been shown to be impaired by the types of stressors that astronauts might expect to encounter (e.g., fatigue, anxiety, constant noise, demanding work shifts). The tests are designed to be self-administered, and the PDA program, MiniCog, is designed to provide immediate performance feedback. The specific tests implemented assess cognitive functioning in the domains of perceptual/motor control, attention (vigilance, divided attention, and filtering), verbal and spatial working memory, cognitive set switching, and verbal and spatial problem solving.		

Task Description:	The tool box developed here has three general applications: First, it will provide a practical means for assessing the effects of various stressors, environmental or internal, on key cognitive processes. For example, it could be used to assess the effects of fatigue, circadian rhythm disruption, mood disorders, exercise, drug use, or diet on cognitive processing. Second, the test battery can be used to assess the effectiveness of "countermeasures," such as specific training to manage fatigue. The tests can be administered before and after such training, which will provide a measure of how the training affects the basic information processes of interest. Finally, the tasks could be used by people "on the job" (e.g., astronauts in space) or others to inform themselves about the current state of their cognitive processes. For example, before EVA to repair an external part, an astronaut might find it useful to know how effectively she or he was reasoning logically and spatially. If one was not doing as well as usual, someone else might go, one might go later, or one might go but make a concerted effort to be very careful. We have finished data collection for an individual differences study using performance on MQAB as a predictor of performance on more complex and longer tasks. We are now analyzing data from that study. A study of the effects of fatigue on performance and the potentially ameliorating effect of caffeine is in the data collection phase. MQAB is also being used in several collaborations, including those with NSBRI team members, Philip Lieberman and Judith Orasanu. We plan to continue this research in the upcoming year, as well as to implement new validation studies, and to begin a collaboration with our NSBRI team leader, David Dinges; finally, we will continue to improve the MiniCog application itself.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The three general applications of this toolbox have "earth" benefits as well as "space" benefits: First, the toolbox will provide a practical means for assessing the effects of various stressors (such as fatigue, exercise, drug use, or diet) on cognitive processing. While we expect some of those variables (e.g., fatigue, certain types of drugs) to have a significant impact on astronauts, they also affect people on earth (e.g., pilots on long flights suffer from irregular sleep schedules; a large percentage of the population takes psychoactive drugs; dieting and exercise are major concerns to many Americans). Second, the tasks can be used to assess the effectiveness of "countermeasures," such as training or drugs to manage fatigue. Just as the same stressors may affect people on the ground as well as the astronauts, the same countermeasures may be useful for the earthbound population as well as those in spaceflight. Finally, the tasks could be used by people "on the job" (e.g. truck drivers) to inform themselves about the current state of their cognitive processes. For example, when truck drivers "weigh in" after many hours of travel, they can run through the battery of tests and make sure that they are still mentally alert. Depending on their scores, they may want to rest before going back on the road, or they may eat a meal or drink coffee, or engage in other activities to reduce the danger of driving. Other researchers will be able to use our test battery to assess cognitive performance quickly and under a variety of laboratory and field conditions; they may also use the MiniCog platform to develop their own psychological experiments. Both our test battery and the MiniCog application have an advantage over standard task batteries and many typical psychological scripting programs in that the tests are brief and the method of administration is compact, portable, and fairly inexpensive. This could make MiniCog and MQAB practical in a wide range of settings where there are questions of neurocognitive capability.
Task Progress:	We have had two types of milestones in the past year: those pertaining to the development of our MiniCog program and those pertaining to our validation of the cognitive tests (MQAB) delivered by the program. Regarding the program itself: (1) We have updated the MiniCog application by improving the user interface to allow more information to be collected prior to the start of a testing session and to make navigation through the program simpler. (2) The software was upgraded to run on newer Palms, and the stimuli have been re-created at a higher resolution to match the capabilities of the new hardware. (3) The data backup method is now more secure. (4) There is more flexibility in specifying the timing of stimulus presentation. Regarding our use of the tests: (1) We have finished data collection for an individual differences study using performance on MQAB as a predictor of performance on more complex and longer tasks. We are now analyzing data from that study. (2) A study of the effects of fatigue on performance, and the potentially ameliorating effect of caffeine, is in the data collection phase. (3) MQAB is also being used in several collaborations, including those with NSBRI team members Philip Lieberman and Judith Orasanu.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Articles in Peer-reviewed Journals	Shephard, J. M. and Kosslyn, S. M. "The MiniCog Quick Assessment Battery: Developing a Aviation Space and Environmental Medicine" Jan-2005
Awards	" Honorary Doctor of Science, University of Caen, Basse-Normandie (France) [2003] " Jan-2005
Awards	" John Simon Guggenheim Memorial Fellowship [2004] " Jan-2005
Patents	not yet assigned Patent Jan-2005 Kosslyn, S. M. and Shephard, J. M. "A System and Method for On-site Cognitive Efficacy Assessment "
Presentation	Kosslyn, S. M. "MiniCog: A 'Blood Pressure Cuff' for the Mind " Jan-2003
Presentation	Shephard, J. M. and Kosslyn, S. M. "Quick Assessment of Basic Cognitive Function: 'Blood Pressure Cuffs' for the Mind (2004) " Jan-2004
Presentation	Shephard, J. M., Chen, J., Kho, S., and Kosslyn, S. M. "Quick Assessment of Basic Cognitive Function: 'Blood Pressure Cuffs' for the Mind (2002) " Jan-2002